AI ENGINEER ROADMAP - 40 WEEKS

4 hrs/day, Mon-Fri | Python -> DSA -> ML basics -> LLM apps -> RAG -> Agents -> Fine-tuning -> Production -> Interview sprint

Start date: ___ / ___ / _____

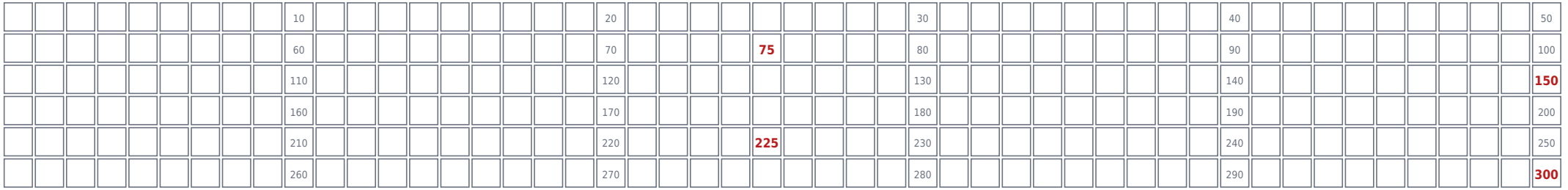
Target finish: ___ / ___ / _____

PHASE 1 (W1-6) **Python, Software Eng. & FastAPI**	**Learn:** Python (OOP, typing, generators, decorators, asyncio), pytest, Pydantic, HTTP, FastAPI, SQL/Postgres, SQLAlchemy, auth, Docker, Redis, CI	MILESTONE ☐ **P1 Prod-style FastAPI service**	weeks done: W1 W2 W3 W4 W5 W6
PHASE 2 (W7-10) **Math, ML & Deep Learning basics**	**Learn:** Linear algebra, calculus, probability, NumPy/Pandas, classical ML, PyTorch, micrograd, tokenization, attention, build a tiny GPT	MILESTONE ☐ **P2 Tiny GPT + ML notes**	weeks done: W7 W8 W9 W10
PHASE 3 (W11-14) **LLM App Engineering**	**Learn:** LLM APIs, prompting, structured outputs, streaming, tool calling, memory, safety, evals 101, tracing, deploy a chat service	MILESTONE ☐ **P3 LLM chat service + evals**	weeks done: W11 W12 W13 W14
PHASE 4 (W15-20) **RAG & Vector Databases**	**Learn:** Embeddings, parsing, chunking, FAISS/Chroma/pgvector/Qdrant, hybrid search, reranking, RAGAS, LangChain v1, ACLs, caching, capstone	MILESTONE ☐ **P4 Production RAG assistant**	weeks done: W15 W16 W17 W18 W19 W20
PHASE 5 (W21-26) **Agents: LangChain, LangGraph, MCP**	**Learn:** Agent loop from scratch, LangChain create_agent + middleware, LangGraph, multi-agent, memory, agent evals, MCP, security, capstone	MILESTONE ☐ **P5 Agent product**	weeks done: W21 W22 W23 W24 W25 W26
PHASE 6 (W27-30) **Hugging Face, Fine-tuning & Serving**	**Learn:** Hugging Face, Trainer, embedding fine-tune, SFT with LoRA/QLoRA, quantization, vLLM, routing, cost modelling, mini-project	MILESTONE ☐ **P6 Fine-tuned + served model**	weeks done: W27 W28 W29 W30
PHASE 7 (W31-34) **Production, Cloud & Flagship Project**	**Learn:** Evals in CI, error analysis, observability, guardrails, AWS deploy, Terraform, load testing, flagship project, portfolio, resume	MILESTONE ☐ **P7 Flagship AI product**	weeks done: W31 W32 W33 W34
PHASE 8 (W35-40) **Interview Sprint (DSA, LLD, HLD, mocks)**	**Learn:** LLD + patterns, system design, AI system design, ML, LLM, Python and SQL question drills, company-specific prep, 3 mock loops, referrals	MILESTONE ☐ **3 mocks + referrals sent**	weeks done: W35 W36 W37 W38 W39 W40

Tick each week box when the weekly deliverable is DONE (not just 'read'). If you slip, extend the week; never skip the deliverable. DSA runs every day alongside all phases.

DSA TRACKER | DAILY RHYTHM | NON-NEGOTIABLES

PROBLEMS SOLVED - colour in one box per problem you can re-solve cold (target 300)



Milestones: [ ] 75 (W10) [ ] 150 (W20 checkpoint) [ ] 225 (W30) [ ] 300 (W34) Mock interviews done: [ ][ ][ ][ ][ ][ ]

DAILY RHYTHM (4h, Mon-Fri)

C - DSA (1h; 2h in W35-40) First, while fresh. 25-30 min try, then solution, then notes. Log fails.
A - Learn (2h) One resource, deep focus, notes in your own words.
B - Build (1h) Apply it today. Commit to GitHub daily.
+5 min Tick the day's A/B/C boxes, write one line in the daily log.
Saturday 2-3h optional: review, catch-up, project polish.
Sunday Off. Sleep and rest are part of the plan.

NON-NEGOTIABLES

1. Code every day, even on bad days (minimum: 1 DSA problem + 30 min build).
2. Understand, then type. Do not copy-paste tutorials.
3. Redo failed DSA problems after 3 days, then 10 days (spaced repetition).
4. Each phase ends with a public repo + README + demo.
5. Measure everything: evals, latency, cost. Put numbers on your resume.
6. Behind schedule? Extend the week. Do not skip the deliverable.
7. Write 5 blog posts. Teaching is the best test of understanding.

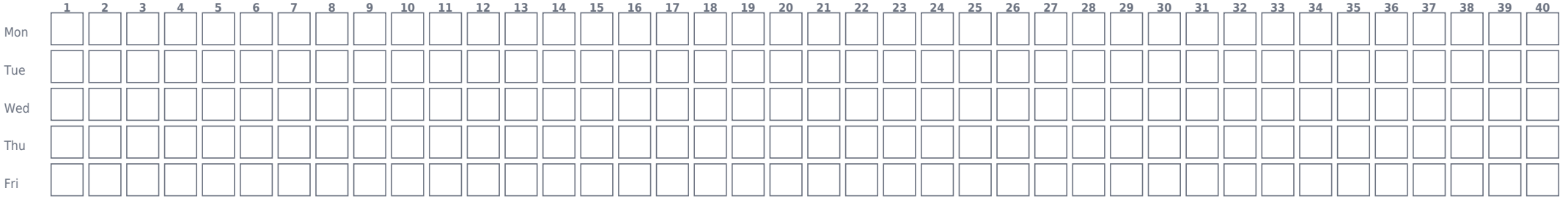
PROJECTS & PUBLIC PROOF

- ☐ P1 FastAPI service (W6)
- ☐ P2 Tiny GPT (W10)
- ☐ P3 LLM chat service (W14)
- ☐ P4 RAG assistant (W20)
- ☐ P5 Agent product (W26)
- ☐ P6 Fine-tuned model (W30)
- ☐ P7 Flagship product (W34)

Blog posts: ☐ ☐ ☐ ☐ ☐

Resume v1 (W34): [ ]
30 target cos list: [ ]
Referral asks sent: ___ / 10+
Applications sent: ___

CONSISTENCY GRID - shade a box for every weekday you finish all 4 hours (40 weeks x 5 days = 200 boxes)



Companies in mind: Google, Meta, Microsoft, Amazon, Zomato, Blinkit, Swiggy, Flipkart, Uber, Atlassian. Their loops weight DSA, system design, LLD and behavioral rounds heavily.